13. Out of 750 families with 4 children each, how many families would be expected to have (i) atleast one boy (ii) atmost 2 girls (iii) and children of both sexes? Assume equal probabilities for boys and girls.

14. Forty percent of business travellers carry a laptop. In a sample of 15 business travelers,

    (i) what is the probability that 3 will have a laptop?

    (ii) what is the probability that 12 of the travelers will not have a laptop?

    (iii) what is the probability that atleast three of the travelers have a laptop?

15. A pair of dice is thrown 4 times. If getting a doublet is considered a success, find the probability of 2 successes.

16. The mean of a binomial distribution is 5 and standard deviation is 2. Determine the distribution.

17. Determine the binomial distribution for which the mean is 4 and variance 3. Also find P(X=15)

18. Assume that a drug causes a serious side effect at a rate of three patients per one hundred. What is the probability that atleast one person will have side effects in a random sample of ten patients taking the drug?

19. Consider five mice from the same litter, all suffering from Vitamin A deficiency. They are fed a certain dose of carrots. The positive reaction means recovery from the disease. Assume that the probability of recovery is 0.73. What is the probability that atleast 3 of the 5 mice recover.

20. An experiment succeeds twice as often as it fails, what is the probability that in next five trials there will be (i) three successes and (ii) at least three successes

7.1.2 Poisson Distribution

Poisson distribution was derived in 1837 by a French Mathematician Simeon D. Poisson. If n is large, the evaluation of the binomial probabilities can involve complex computations, in such a case, a simple approximation to the binomial probabilities could be use. Such approximation of binomial when n is large and p is close to zero is called the Poisson distribution.

Poisson distribution occurs when there are events which do not occur as a definite number on trials but an events occurs rarely and the following examples may be analysed:

(i) Number of bacteria in one cubic centimeter.

(ii) Number of printing mistakes per page in a text book

(iii) the number of alpha particles emitted by a radioactive substance in a fraction of a second.

(iv) Number of road accidents occurring at a particular interval of time per day.

Number of lightnings per second.

Poisson distribution is a limiting case of binomial distribution under the following conditions :

(i) n, the number of trials is indefinitely large i.e $n \to \infty$.

(ii) p, the constant probability of success in each trial is very small, i.e. $p \to 0$

(iii) $np = \lambda$ is finite. Thus $p = \frac{\lambda}{n}$ and $q = 1 - \left(\frac{\lambda}{n}\right)$ where λ is a positive real number.

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